# Python ALA 2 = Accrssing List elements in python

1. What is a List?
A list is a collection of elements stored in a single variable.

my_list = [10, 20, 30, 40, 50]

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**2. Access Using Index (Most Important)**
👉 **Python uses 0-based indexing**

**Example**

```
print(my_list[0])  # 10
print(my_list[1])  # 20
print(my_list[2])  # 30
```

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**3.Negative Indexing**
👉 Access elements from the end

**Example:**
```
print(my_list[-1])  # 50 (last element)
print(my_list[-2])  # 40
```

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**4.Slicing (Access Multiple Elements)**
Example
```
print(my_list[1:4])  # [20, 30, 40]
```

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5. Access Nested List

**Example:**
```
nested = [[1, 2], [3, 4]]

print(nested[0][1])  # 2
```

6.Modify Elements

**Example**

**Modify Elements**

```
my_list[1] = 25
print(my_list)  # [10, 25, 30, 40, 50]
```

**7. What is time complexity of index access?**
**Answer:** O(1) (very fast)

**8. Why does Python use 0-based indexing?**
**Answer:**
Because it matches memory address calculation and makes indexing faster and simpler
**EXAMPLE**

```
my_list = [10, 20, 30, 40, 50]
print(my_list[1:4])
```

**9. Important Interview Points**

- Lists are **mutable**
- Index starts from **0**
- Negative indexing supported
- Slicing returns a **new list**
- Index access is **fast (O(1))**

**10. Can slicing change original list?**
No
Slicing creates a **new list**.
**EXAMPLE**

```
a = [1,2,3,4]
b = a[1:3]

print(b)  # [2,3]
```

**Complete Example**

```
my_list = [10, 20, 30, 40, 50]

print("First element:", my_list[0])
print("Last element:", my_list[-1])
print("Slice:", my_list[1:4])

for item in my_list:
    print(item)
```